Testing and Testable Design

(Kỹ thuật DFT và kiểm thử)

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Vị trí việc làm

6. DFT Engineer

Faraday Technology Corporation | Thành phố Hồ Chí Minh

You must create an Indeed account before continuing to the company website to apply

Apply on company site

Full job description

Responsibilities

- Define DFT specification, finalize DFT strategy to achieve the testing target
- Perform implementation and verification DFT circuit (MBIST/SCAN/BSCAN/IP-Test)
- Define DFT SDC (timing constraint) strategy, collaborate with Backend engineer to validate DFT circuit & SDC
- Tester pattern generation and silicon debug analysis/diagnosis
- Work as project leader/block coordinator to control project schedule and support project members to ensure T/O on-time

Requirements

- Familiar with Tessent/TMAX, synthesis/simulation/formal verification tools
- Excellent team and interpersonal skills
- BS or MS degree in EE or CS related
- Good English communication skills

DFT Staff Engineer

Synopsys | Thành phố Hồ Chí Minh

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- Design for Test Staff Engineer
- DFT Implementation Staff Engineer

We Are

Synopsys is the leader in engineering solutions from silicon to systems, enabling customers to rapidly innovate AI-powered products. We deliver industry-leading silicon design, IP, simulation and analysis solutions, and design services. We partner closely with our customers across a wide range of industries to maximize their R&D capability and productivity, powering innovation today that ignites the ingenuity of tomorrow.

You Are

You have spent years thinking about how chips break and how to catch those failures before they reach a customer. Testability is not an afterthought for you, it is a design decision that happens early, and you see the downstream consequences of every scan chain stitch, every ATPG pattern, every timing constraint you write. When something does not simulate cleanly, you dig in, trace it back, and figure out what actually went

Tools

- Lecture notes from the LMS
- Textbook
- Evaluation
- Software

Textbook

- **Giáo trình chính**:

[1] Laung-Terng Wang, Cheng-Wen Wu, Xiaoqing Wen, *VLSI Test Principles and Architectures: Design for Testability*, Morgan Kaufmann, 2006.

[2] Adam Teman, *Design For Test (DFT)*, Bar-Ilan University, 2020.

- Tài liệu tham khảo:

[3] Alfred Crouch, *Design-for-Test for Digital IC's and Embedded Core Systems*, Prentice Hall, 1999.

[4] Nur Touba, *Electronic Design Automation for Integrated Circuits Handbook – DFT Section*, 2006.

[5] IEEE 1149.1 Standard Documentation, 2013.

[6] https://enicslabs.com/

Chuẩn đầu ra của học phần (CLOs)

CLOs	Description (After completing this module, learners will be able to)	PLOs / PI(s)	Level
CLO1	Explain and analyze the principles of digital integrated circuit testing, fault models, VLSI testing procedures, and evaluation metrics such as yield, reliability, fault coverage, and test cost.	PI1.1	4
CLO2	Analyze and evaluate the testability of digital circuits through metrics such as controllability, observability, SCOAP, and techniques for improving testability.	PI2.1	4
CLO3	Design and implement Design for Testability (DFT) techniques, including scan design, scan chain, ATPG, BIST logic, and evaluate the effectiveness of the DFT process for VLSI/SoC systems.	PI3.2	4
CLO4	Capable of identifying the test requirements of a VLSI/SoC system and proposing suitable Design-for-Testability (DFT) solutions to improve the testability and reliability of the design.	PI4.1	4
CLO5	Capable of designing, implementing, and evaluating DFT processes for digital circuits, including scan insertion, ATPG, BIST logic, and fault coverage analysis to meet design testing requirements.	PI4.2	4
CLO6	Effectively utilize EDA tools, HDL/scripting languages, and AI tools to support testability analysis, DFT flow implementation, test pattern generation (ATPG), simulation, and test result evaluation.	PI6.2	4

Content

week	content
1	Chapter 1: Introduction to VLSI Testing and Manufacturing
2	Chapter 2: Fundamentals of Fault Modeling and Structural Testing
3	Chapter 3: Design for Testability and Scan Design
4	Chapter 4: Scan Architectures, Design Rules
5	Review Questions and Hands-on Exercises
6	Mid-term Examination
7	Project Preparation and Consultation
8	Final project seminar (session1)
9	Final project seminar (session 2)

Evaluation

Assessment Component	Weight (%)
Attendance	10%
In-Class Quizzes	10%
Midterm Exam	30%
Final Project (Presentation, Report, and Video Demonstration)	50%
Bonus Points (Exercises, Class Participation, Mini Project)	Up to +10%

Additional Topics

3–4 students / group / 1 topic

1 group / 1 topic

Presentation in class (~20–30 minutes)

1. Career Opportunities in IC Design in Vietnam

- Major IC companies in Ho Chi Minh City, Da Nang, and Can Tho
- Career paths (RTL Design, Verification, DFT, Physical Design, Analog, Embedded)
- Required skills and salary overview

Additional Topics

2. Design for Testability (DFT) Engineer Career

- Roles and responsibilities
- Typical DFT flow
- Required knowledge and EDA tools

3. How to Write an IC Design CV

- Resume structure
- Highlighting technical projects
- GitHub and portfolio
- Common mistakes

4. Future Trends in Semiconductor Industry

- AI
- Chiplets
- Advanced Packaging
- Quantum Computing